

SOIL ANALYSIS

17

Both planning and evaluation of irrigation project requires relevant and sufficient information of soil in the irrigation command. Soil type and its characteristic in the irrigation command area are the most important governing parameters for selecting a crop and water management practices as well as in deciding the irrigation application method. Water losses in the field are also different for different soil characteristics. Looking into this, this chapter addresses the physical characteristics of the soil, including the soil-water constants with a detailed protocol for analyzing the same.

17.1 SOIL AND ITS PHYSICAL PROPERTIES

Soil is a typical heterogeneous multiphase porous system, which, in its general form, contains three natural components: (1) the solid phase of the soil matrix (formed by mineral particles and solid organic materials); (2) the liquid phase, which is often represented by water and which could more properly be called the soil solution; and (3) the gaseous phase, which contains air and other gases. Practically speaking, a soil particle must pass through a 2-mm sieve to be called soil particle. Soil particles are either organic or inorganic. For optimum plant growth, it is essential that the soil provides a favorable physical environment for root development that meets the plant needs for water, nutrients, and anchorage. Other essential conditions include aeration, optimum soil temperature, seed emergence, and root penetration facility (low mechanical impedance).

The physical properties and chemical composition largely determine the suitability of a soil for its planned use and the management requirements to keep it productive. The physical characteristics of soil largely determine the movement of water into and through the soil matrix (the infiltration rate at the soil surface, and the way water is redistributed throughout the root zone, under the action of gravity and soil suction forces), and the amount of soil water available to plants (the soil's water holding capacity).

Soil texture is determined by analyzing the proportions of sand, silt, and clay particles. Soil structure is difficult to measure, but for a particular soil type, its bulk density may give some information on the volume of pores in the soil.

The behavior of the soil when wetted should also be assessed. In particular, some soils have a tendency for slaking (a breakdown of the structure of soil crumbs when wetted) or dispersion (where the clay particles are released into suspension). These characteristics are undesirable and contribute to problems, such as surface crusting and poor internal drainage, which in turn result in poor crop establishment and performance and ultimately difficult irrigation management. Slaking will be observed if the crumbs appear to disintegrate and form a layer at the bottom of the dish. Dispersion will result in the water becoming cloudy. These characteristics are partly affected by the chemical analysis of water,

particularly by the presence of sodium, and the sodium absorption rates of the soil and water. Consequently, dispersion and slaking should be assessed both with distilled water and with irrigation water.

17.1.1 SOIL TEXTURE

This term refers to the relative proportion of sand, silt, and clay in the soil. Each of these components is defined by its particle size (sand 0.02–2.0 mm, silt 0.02–0.002 mm, clay less than 0.002 mm). The soil texture is more correctly referred to as the soil's particle size distribution.

The main soil types can be classified according to their texture. Fig. 17.1 shows a commonly accepted classification using the textural triangle derived in the United States. The commonly used classification proposed by the United States Department of Agriculture and the International Soil Science Society is given in Table 17.1.

To determine soil texture, a sample of soil is oven dried, weighed, crushed, and passed through a set of sieves of particular mesh sizes. The amount of material retained by each sieve is weighed. Silt and clay (the fines) both pass the smallest sieve and are collected in a tray and are separated from each other in a different way. The sand and coarse fractions are weighed and recorded as a percentage of the initial weight.

17.1.2 SOIL STRUCTURE

Soil structure refers to the grouping of soil particles (sand, silt, clay, organic matter, and fertilizers) into porous compounds. These are called aggregates. Soil structure also refers to the arrangement of these aggregates separated by pores and cracks (Fig. 17.2).

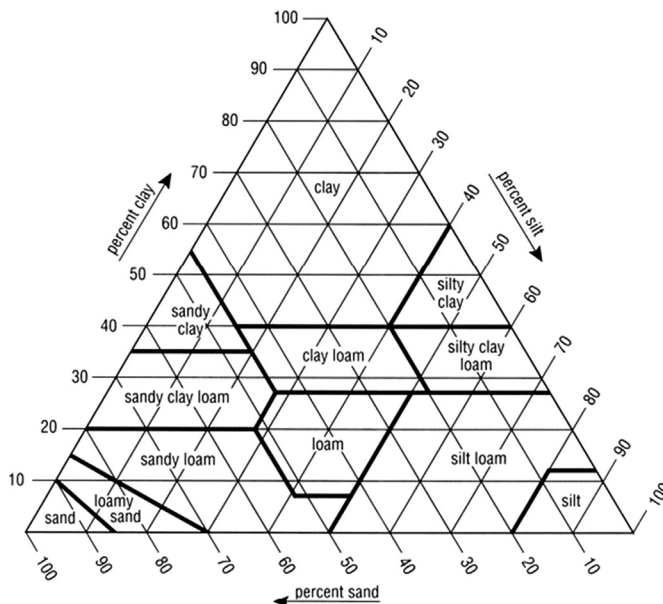


FIGURE 17.1

The United States Department of Agriculture particle size distribution classifications.

Table 17.1 Textural Classification of the USDA and ISSS

Fraction	Particle Diameter (mm)	
	USDA	ISSS
Gravel	>2.0	>2.0
Very fine sand	1.0–2.0	—
Coarse sand	0.5–1.0	0.2–2.0
Medium sand	0.25–0.5	—
Fine sand	0.1–0.25	0.02–0.20
Very fine sand	0.05–0.10	—
Silt	0.002–0.050	0.002–0.02
Cay	<0.002	<0.002

ISSS, *International Soil Science Society*; USDA, *United States Department of Agriculture*.

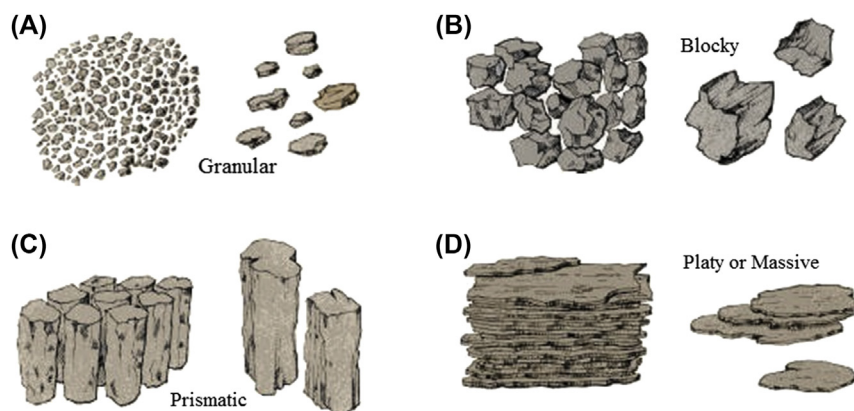
**FIGURE 17.2**

Soil structure.

Soil structure will also influence water movement down the profile after infiltration. A well-structured soil will drain relatively quickly after heavy rain or irrigation and so minimize the problem of temporary water logging. Although draining freely, a heavy-textured but well-structured soil will still hold adequate quantities of water available to plant roots. Poorly structured soil may inhibit plant growth and development by being waterlogged in wet conditions, thereby restricting the development of the plant root system.

The basic types of aggregate arrangements are: granular, blocky, prismatic, and massive structures (Fig. 17.3). When present in the topsoil, a massive structure blocks the entrance of water, and seed germination is difficult due to poor aeration. On the other hand, if the topsoil is granular, the water enters easily and seed germination is better.

In a prismatic structure, the movement of water in the soil is predominantly vertical and therefore the supply of water to the plant roots is usually poor. Unlike texture, soil structure is not permanent. By means of cultivation practices (ploughing, ridging, etc.), the farmer tries to obtain a granular topsoil structure for agricultural fields.

**FIGURE 17.3**

Basic types of aggregate arrangements. (A) Granular, (B) blocky, (C) prismatic, and (D) massive.

Table 17.2 Soil Depth Classification	
Class	Depth (Topsoil + Subsoil), cm
Very shallow	<25
Shallow	25–50
Moderately deep	50–75
Deep	≥75

17.1.3 SOIL DEPTH

Soil depth indicates how deep (top to bottom) the topsoil plus the subsoil is. The depth can be determined by digging a hole. Soil depth is important for plants because deeper rooting means more soil to exploit for nutrients and water. Greater soil depth can also mean better drainage, as long as there are no restrictive layers in the subsoil. Soils may be classified as being deep or shallow using [Table 17.2](#).

17.1.4 OTHER PHYSICAL CHARACTERISTICS OF SOIL

17.1.4.1 Bulk Density

Soil bulk density is the mass of soil per unit volume. Bulk density provides an estimate of the total water storage capacity, texture, infiltration rate, compactness, or aeration condition. In three-phase (solid, liquid and gas) soil system, soil density can be categorized as soil particle density, bulk or dry density, and wet density.

Within a certain volume of soil, the total mass can be expressed as follows:

$$M = M_s + M_l + M_g \quad (17.1)$$

where M_s is the mass of solid, M_l is the mass of liquid, and M_g is the mass of gases. Since the mass of the gases is negligible as compared to the mass of solid and liquid, the form of the aforementioned relationship can be truncated. Thus Eq. (17.1) will take the following form:

$$M = M_s + M_l \quad (17.2)$$

Similarly, the volume of soil V can be defined as follows:

$$V = V_s + V_l + V_g \quad (17.3)$$

In Eq. (17.3), V_s is the volume of solids, V_l is the volume of liquid, and V_g is the volume of gas.

Considering this concept (Eqs. 17.2 and 17.3), the bulk or dry density is defined as the ratio of mass of solid to the total volume of soil.

$$\rho_b = \frac{M_s}{V} = \frac{M_s}{V_s + V_l + V_g} \quad (17.4)$$

Typical values of the dry or bulk density of most soil vary within the range of 1.1–1.6 g/cm³. An average value would be 1.3 g/cm³ for loamy soil. The bulk density of different textured soil is summarized in Table 17.3.

17.1.4.2 Particle Density

Similar to the bulk density, the particle density is defined as the ratio of mass of solid particle to the volume of solids.

$$\rho_s = \frac{M_s}{V_s} \quad (17.5)$$

Table 17.3 Soil Physical Characteristics of Different Textured Soil			
Property	Texture		
	Sand	Silt	Clay
Bulk density (g/cm ³)	Low (1.0–1.1)	Medium (1.1–1.3)	High (1.3–1.6)
Porosity (%)	Low	Medium	High
Pore size	Large (mostly large pores)	Medium (small pores predominates)	Small (small pores predominates)
Soil particle surface (i.e., surface area for a given volume)	Small	Medium	Large
Water holding capacity	Low	Medium	High
Permeability (mm/d)	Rapid (>10)	Slow to moderate (2–8)	Slow (~2)
Field capacity (i.e., upper limit of plant's available water)	Low (<35% v/v)	Medium (<45% v/v)	High (45–55% v/v)
Wilting point (i.e., lower limit of plant's available water)	Low (<15% v/v)	Medium (<20% v/v)	High (20–30% v/v)

The particle density would be equivalent to the average density of the soil minerals and organic matter. It varies between 2.6 and 2.7 g/cm³.

Example 17.1: A sharp-edged cylinder of 5 cm internal diameter is carefully driven into the soil so that negligible compaction occurs. The length of soil column is 8 cm. The weight of the soil is 280 g. After it is oven dried at 105°C, unless constant weight is achieved, the weight of the soil is 230 g. Determine the following: (1) the moisture percentage on a dry weight basis, (2) the apparent specific gravity of the soil, and (3) the percent moisture on a volumetric basis.

Solution:

Given: internal diameter of the cylinder, $d = 5$ cm

Length of the soil column, $h = 8$ cm

Volume of soil, $V = \frac{\pi}{4}d^2h = 157.08 \text{ cm}^3$

Total weight of the soil, $W = 280$ g

Dry weight of the soil, $W_s = 230$ g

Weight of moisture, $W_l = 280 - 230 = 50$ g

1. Moisture percent on a dry weight basis, $\theta_{(W/W)} = \frac{W_l}{W_s} \times 100 = (50/230) \times 100 = 21.7\%$

2. Apparent specific gravity of the soil, $\gamma_b = \frac{W_s}{V} = 230/157.08 = 1.46 \text{ g/cm}^3$

3. Percent moisture on a volumetric basis $\theta_{(V/V)} = \theta_{(W/W)} \times \gamma_b = 21.7 \times 1.46 = 31.8\%$.

Example 17.2: A cylinder is carefully penetrated into the soil without disturbing the soil. The cross-sectional area of the cylinder is 40 cm² with a soil column of 30 cm. The weight of the soil within the cylinder is 2.0 kg. After oven drying the soil at 105°C for 36–48 h, the weight of the dry soil is 1.65 kg. Determine the following: (1) the current soil moisture percentage on a dry weight basis, (2) the bulk density or apparent specific gravity of the soil, and (3) the percent moisture on a volumetric basis.

Solution:

Given: cross-sectional area of the cylinder, $A = 40 \text{ cm}^2$

Length of the soil column, $h = 30$ cm

Volume of soil, $V = Ah = 1200 \text{ cm}^3$

Total weight of the soil, $W = 2000.0$ g

Dry weight of the soil, $W_s = 1650$ g

Weight of moisture, $W_l = 2000 - 1650 = 350$ g

1. Moisture percent by dry weight basis, $\theta_{(W/W)} = \frac{W_l}{W_s} \times 100 = (350/1650) \times 100 = 21.2\%$

2. Apparent specific gravity of the soil, $\gamma_b = \frac{W_s}{V} = 1650/1200 = 1.38 \text{ g/cm}^3$

3. Percent moisture by volumetric basis $\theta_{(V/V)} = \theta_{(W/W)} \times \gamma_b = 21.2 \times 1.38 = 29.2\%$.

17.1.4.3 Porosity

Porosity refers to the portion of the soil volume that is occupied by the pore-space or voids. It is normally expressed as a percentage of soil volume:

$$\phi = \frac{V_v}{V} \times 100\% = \frac{V_v}{V_v + V_s} \times 100\% \quad (17.6)$$

where, ϕ is the porosity (%), V_v is the volume of voids (cm³), and V is the total volume of soil.

17.1.4.4 Relationship Between Porosity (ϕ), Bulk Density (ρ_b), and Particle Density (ρ_s)

A value for porosity can alternatively be calculated from the bulk density ρ_b and particle density ρ_s as follows:

$$\phi = 1 - \frac{\rho_b}{\rho_s} = 1 - \frac{M_s/V}{M_s/V_s} = 1 - \frac{V_s}{V} = \frac{V - V_s}{V} = \frac{V_v}{V} \quad (17.7)$$

From Eq. (17.7), it is obvious that the bulk density is smaller than the particle density. Typical values of porosity for sandy, loam, and clay textures can range between 30%–40%, 40%–45%, and 45%–55%, respectively.

17.1.4.5 Relationship Between Porosity (ϕ) and Void Ratio (e)

The void ratio is defined as the ratio of volume of voids and the volume of solids present in the soil. It is expressed as follows:

$$e = \frac{V_v}{V_s} \quad (17.8)$$

$$e = \frac{V_v}{V - V_v} = \frac{1}{\frac{V}{V_v} - 1} = \frac{1}{\frac{1}{\phi} - 1} = \frac{\phi}{1 - \phi} \quad (17.9)$$

Alternatively, the porosity can be related with the void ratio as follows:

$$\phi = \frac{V_v}{V} = \frac{V_v}{V_s + V_v} = \frac{1}{\frac{V_s}{V_v} + 1} = \frac{1}{\frac{1}{e} + 1} = \frac{e}{1 + e} \quad (17.10)$$

where e is the void ratio, ϕ is the porosity, V_v is the volume of void space ($V_w + V_a$), V_s is the volume of solids, and V is the total or bulk volume.

Example 17.3: A soil has a real (i.e., particle) density and bulk density of 2.6 g/cm³ and 1.4 g/cm³. Determine the porosity and void ratio of the soil.

Solution:

Given: The particle density of the soil, $\rho_s = 2.6$ g/cm³

The bulk density of the soil, $\rho_b = 1.4$ g/cm³

The porosity of the soil, $\phi = 1 - \frac{\rho_b}{\rho_s} = [1 - (1.4/2.6)] \times 100 = 46.1\%$

The void ratio of the soil, $e = \frac{\phi}{1 - \phi} = \frac{0.461}{1 - 0.461} = 0.855$

17.1.4.6 Water Holding Capacity and Drainability

The field water holding capacity is the amount of water that a soil can hold after being saturated and allowed to drain for a period of 1–2 days. Field water holding capacity is influenced by soil texture, aggregation, organic matter content, and overall soil structure.

It can also be defined as the degree to which a soil is capable of holding or retaining moisture between field capacity and wilting point. It will limit the proportion of rainfall that will be held for subsequent use by the crop.

Drainability refers to the ability of a soil to get rid of excess water, or water in macropores, through downward movement by gravity. Heavy clay soils that have a lot of micropores have a higher water holding capacity, but because they have fewer macropores, their drainage is poor.

17.2 SOIL-WATER CONSTANTS

The soil moisture content indicates the amount of water present in the soil. It is commonly expressed as the amount of water (in millimeters of water depth) present at a depth of 1 m of soil. The soil moisture content can also be expressed in percent of volume. Fig. 17.4 depicts different terms of the soil–water relationship. The physical interpretation of soil moisture status can be visualized from Table 17.4.

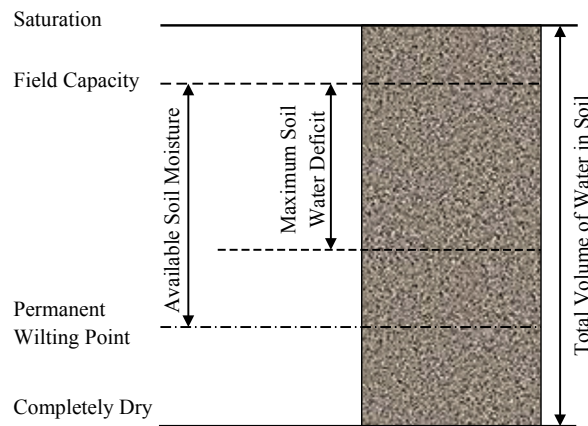


FIGURE 17.4
Soil–water terminologies.

Table 17.4 Soil Moisture Interpretation Chart			
Soil Moisture Deficiency (% of FC)	Moderately Coarse Texture	Medium Texture	Fine and Very Fine Texture
0	Upon squeezing, no free water appears on soil but wet outline of ball is left on hand		
0–25	Forms weak ball, breaks easily when bounced in hand ^a	Forms ball, very pliable, slicks readily ^a	Easily ribbons out between thumb and forefinger ^a
25–50	Will form ball, but falls apart when bounced in hand ^a	Forms ball, slicks under pressure ^a	Forms ball, will ribbon out between thumb and forefinger ^a
50–75	Appears dry, will not form ball with pressure ^a	Crumbly, holds together from pressure ^a	Somewhat pliable, will ball under pressure ^a
75–100	Dry, loose, flows through fingers	Powdery, crumbles easily	Hard, difficult to break into powder
^a Squeeze a handful of soil firmly to make ball test.			

For example, if 1 m^3 of soil (e.g., with a depth of 1 m, and a surface area of 1 m^2) contains 0.20 m^3 of water (e.g., with a depth of 200 mm = 0.20 m and a surface area of 1 m^2), then the soil moisture content in volume percent will be 20%.

17.2.1 FIELD CAPACITY

The term field capacity is interchangeably used with the terms the water holding capacity and water retention capacity. Field capacity is the amount of soil moisture or water content held in soil after excess water has drained away and the rate of downward movement has materially decreased, which usually takes place within 2–3 days after a rain or irrigation in pervious soils of uniform structure and texture. The physical definition of field capacity (θ_{fc}) is the bulk water content retained in soil at -33 J/kg (or -0.33 bar) of hydraulic head or suction pressure.

When irrigation is applied to the soil, all the soil pores get filled with water. After the gravitational drainage, the large soil pores are filled with both air and water, while the smaller pores are still full of water. At this stage, the soil is said to be at field capacity. At field capacity, the water and air contents of the soil are considered to be ideal for crop growth.

17.2.2 SATURATION CAPACITY

During a rain shower or irrigation application, the soil pores will be filled with water. If all soil pores are filled with water, the soil is said to be saturated. There is no air left in the soil. It is easy to determine in the field if a soil is saturated. If a handful of saturated soil is squeezed, some (muddy) water will run between the fingers.

Plants need air and water in the soil. At saturation, no air is present and the plant will suffer. Many crops cannot withstand saturated soil conditions for a period of more than 2–5 days. Rice is one of the exceptions to this rule. The period of saturation of the topsoil usually does not last long. After the rain or the irrigation has stopped, part of the water present in the larger pores will move downward. This process is called drainage or percolation.

The water drained from the pores is replaced by air. In coarse-textured (sandy) soils, drainage is completed within a period of a few hours. In fine-textured (clayey) soils, drainage may take some (2–3) days.

17.2.3 PERMANENT WILTING POINT

The permanent wilting point is the point when there is no water available to the plant. The permanent wilting point depends on plant variety, but it is usually around 1500 kPa (15 bars). At this stage, the soil still contains some water, but it is difficult for the roots to extract from the soil. Nearly 15 bars of tension is required to extract water by the plants. At this limit, if no additional water is supplied to the soil, most of the plants die.

The moisture content at the permanent wilting point varies with soil texture. Fine-textured soils retain higher amounts of water ($\sim 26\%–32\% \text{ v/v}$) than the coarse textured soils ($10\%–15\% \text{ v/v}$) at the permanent wilting point.

17.2.4 AVAILABLE WATER CAPACITY

The soil can be compared to a water reservoir for the plants. When the soil is saturated, the reservoir is full. However, some water drains rapidly below the root zone before the plant can use it. When this

water has drained away, the soil is at field capacity. The plant roots draw water from what remains in the reservoir. When the soil reaches permanent wilting point, the remaining water is no longer available to the plant.

The amount of water actually available to the plant is the amount of water stored in the soil at field capacity minus the water that will remain in the soil at permanent wilting point. The available water content depends greatly on the soil texture and structure.

The moisture at the available water capacity (AWC) is expressed as follows:

$$\theta_{AWC} = \theta_{FC} - \theta_{PWP} \quad (17.11)$$

where θ_{AWC} is the maximum available moisture content (% v/v), θ_{FC} is the moisture content at field capacity (% v/v), and θ_{PWP} is the moisture content at permanent wilting point (% v/v). The available water content can be given as follows:

$$AWC = \frac{\theta_{FC} - \theta_{PWP}}{100} \times D \quad (17.12)$$

In Eq. (17.12), AWC is the available water holding capacity or maximum water availability for the soil (cm), and D is the root zone depth of the soil (cm). If the moisture content at field capacity and permanent wilting point is measured on a weight basis (i.e., % w/w), then AWC is estimated as follows:

$$AWC = \frac{\theta_{FC(w/w)} - \theta_{PWP(w/w)}}{100} \times \gamma_b \times D \quad (17.13)$$

where γ_d is the apparent specific gravity or the bulk density of the soil (g/cm³). For layered soil, this relationship takes the following form:

$$AWC = \sum_{i=1}^n \frac{\theta_{FC_i} - \theta_{PWP_i}}{100} \times D_i \quad (17.14)$$

where subscript i denotes the layer from 1 to the n th layer.

For various soil textures, the maximum available moisture is given in Table 17.5.

17.2.5 PRESENTLY AVAILABLE SOIL MOISTURE

The presently available soil moisture ($PASM$) is the moisture currently (i.e., at present state of the crop and soil) available to the plant. Mathematically, it is expressed as follows:

$$\theta_{PASM} = \theta_0 - \theta_{PWP} \quad (17.15)$$

where θ_{PASM} is the $PASM$ content (% v/v) and θ_0 is the current soil moisture content (% v/v).

It can be presented in the depth terms through the following equation:

$$PASM = \frac{\theta_0 - \theta_{PWP}}{100} \times D \quad (17.16)$$

Table 17.5 Maximum Available Soil Moisture for Various Soil Textures		
Soil Texture	Available Water Capacity (cm/m)	
	Range	Average
Coarse sand	2.08–6.25	4.17
Fine sand	6.25–8.33	7.29
Loamy sand	9.17–10.00	9.58
Sandy loam	10.42–11.67	11.04
Fine sandy loam	12.50–16.67	14.58
Silt loam	16.67–20.83	18.75
Silty clay loam	15.00–16.67	15.83
Silty clay	12.50–14.47	13.33
Clay	10.00–12.50	11.25

where $PASM$ is presently available soil moisture to the plant (cm) and D is the soil depth up to root zone (cm). For layered soil, Eq. (17.16) can be reformed as follows:

$$PASM = \sum_{i=1}^n \frac{\theta_{0i} - \theta_{PWPi}}{100} \times D_i \quad (17.17)$$

where D_i is the depth of the i th layer (cm) and n is number layers in the effective rooting depth.

17.2.6 DEPLETION OF AVAILABLE SOIL MOISTURE

The percentage depletion of available soil water is the lowering of current state of soil moisture from field capacity with respect to the theoretical maximum possible available soil moisture. It is expressed as follows:

$$\text{Depletion, \%} = \frac{\theta_{FC} - \theta_0}{\theta_{FC} - \theta_{PWP}} \times 100 \quad (17.18)$$

For layered soils, the average depletion of available soil moisture can be estimated from Eq. (17.19).

$$\text{Average Depletion, \%} = \frac{1}{n} \sum_{i=1}^n \frac{\theta_{FCi} - \theta_{0i}}{\theta_{FCi} - \theta_{PWPi}} \quad (17.19)$$

In Eq. (17.19), θ_{0i} is the current soil moisture of the i th layer (% v/v).

Example 17.4: A soil sample has been extracted from the field which represents the uniform moisture within the rooting depth of 80 cm. The volume of sample is 1000 cm³ with a mass of 1750 g.

The same soil sample is completely wetted and drained gravitationally for 2–3 days. The mass of the soil sample after draining is 1900 g. After air drying the sample the mass of soil is 1650 g. Finally, the sample was oven dried at 105°C for more than 30 h, and the mass of sample was recorded as 1500 g.

Calculate (1) bulk density of soil, (2) current soil moisture content by weight, (3) current moisture content by volume, (4) moisture content at field capacity by volume, (5) moisture content at permanent wilting point by volume, (6) maximum water availability, (7) *PASM*, and (8) depletion of available soil moisture.

Solution:

Given: Volume of soil, $V = 1000 \text{ cm}^3$

Weight of the soil, $W_t = 1750.0 \text{ g}$

Dry weight of the soil, $W_s = 1500 \text{ g}$

Weight of soil at field capacity = 1900 g

Weight of soil at permanent wilting point = 1650 g

Depth of rooting, $D = 80 \text{ cm}$

1. Bulk density of soil or apparent specific gravity of the soil:

$$\gamma_b = \frac{W_t}{V} = 1750/1000 = 1.75 \text{ g/cm}^3$$

2. Current soil moisture content by weight, θ_0 (% w/w):

$$\theta_{0(w/w)} = \frac{W_t - W_s}{W_s} \times 100 = \frac{1750 - 1500}{1500} \times 100 = 16.66\%$$

3. Current soil moisture content by volume, θ_0 (% v/v):

$$\theta_{0(v/v)} = \theta_{0(w/w)} \times \gamma_b = 16.66 \times 1.75 = 29.16\%$$

4. Moisture content at field capacity by volume:

$$\theta_{FC(v/v)} = \theta_{FC(w/w)} \times \gamma_b = \frac{W_{FC} - W_s}{W_s} \times 100 \times \gamma_b$$

$$\theta_{FC(v/v)} = \frac{1900 - 1500}{1500} \times 100 \times 1.75 = 46.67\%$$

5. Moisture content at permanent wilting point by volume:

$$\theta_{PWP(v/v)} = \theta_{PWP(w/w)} \times \gamma_b = \frac{W_{PWP} - W_s}{W_s} \times 100 \times \gamma_b$$

$$\theta_{PWP(v/v)} = \frac{1650 - 1500}{1500} \times 100 \times 1.75 = 17.5\%$$

6. Maximum water availability or water holding capacity, *AWC*:

$$AWC = \frac{\theta_{FC(v/v)} - \theta_{PWP(v/v)}}{100} \times D = \frac{46.67 - 17.5}{100} \times 80 = 23.34 \text{ cm}$$

7. PASM:

$$PASM = \frac{\theta_{0(v/v)} - \theta_{PWP(v/v)}}{100} \times D = \frac{25 - 15}{100} \times 80 = 8 \text{ cm}$$

8. Depletion of available soil moisture

$$\text{Depletion, \%} = \frac{\theta_{FC} - \theta_0}{\theta_{FC} - \theta_{PWP}} \times 100 = \frac{40 - 25}{40 - 15} \times 100 = 60\%$$

17.3 PROTOCOL FOR SOIL MOISTURE ANALYSIS USING GRAVIMETRIC METHOD

17.3.1 PROTOCOL FOR THE DETERMINATION OF BULK DENSITY OF SOIL

The following procedure can be used to estimate the bulk density of soil:

1. Identify three sample locations in the field representative of the overall field.
2. Collect soil samples from three depths using the sharp-edged cylindrical auger of 5 cm internal diameter. The auger should be driven carefully into the soil so that negligible compaction occurs. The length of soil column should at least be 10 cm (Fig. 17.5).
3. Take the weight of the sample in the field.
4. Once the sample is weighed it will be oven dried at 105°C unless constant weight is achieved. It results in dry weight of the soil.
5. The bulk density is estimated by dividing the dry weight of the soil material (W_d) by the volume of soil (V) or inner volume of auger.
6. The data will be collected and entered in the format given in Table 17.6.



FIGURE 17.5

Soil sample taken by the auger.

Table 17.6 Protocol for Determination of BD of Soil of Field Plot

Field plot location:Lat,Long

Village:

Name or Outlet ID and RD:

Name of Minor/Distributary/Main canal:.....

Chak No.:.....

Khasra No.:.....

General Soil Characteristics: (i) Type:, (ii) Soil depth:.....

General Topography:.....

General Weather Condition During Sampling:

Date of Sampling:

Sample IDs:

Plot Location No.	Soil Depth From Which the Sample Taken (cm)												Average BD (g/cm ³)
	0–20 cm				20–50 cm				50–100 cm				0–100 cm
	W_i (g)	V (cm ³)	W_d (g)	BD (g/cm ³)	W_i (g)	V (cm ³)	W_d (g)	BD (g/cm ³)	W_i (g)	V (cm ³)	W_d (g)	BD (g/cm ³)	
1													
2													
3													
4													

BD = W_d/V (g/cm³); V = volume of sample (cm³) $\left[V = \frac{\pi}{4}d^2h \right]$; W_d = Weight of oven dried sample at 105°C until the weight becomes constant; W_i = initial weight of the sample taken at the time of sample (g).
 BD, bulk density.

17.3.2 PROTOCOL FOR THE DETERMINATION OF VARIOUS WATER CONSTANTS

The following procedure can be used to estimate the various water constants like field capacity and permanent wilting point of the soil:

1. Identify the sampling locations in the field plot under study. At least three representative locations are to be identified.
2. At each location, three soil samples will be taken from various depths without disturbing the soil.
3. At the time of sampling, carefully measure the volume (V) as well as the current weight (W_i) of the soil sample.
4. Take the sample to the laboratory and measure the weight at different conditions:
 - a. Completely wet the soil and drain it for 2–3 days and take the weight of the sample, which is the weight of the soil at field capacity (W_{FC}).
 - b. Air dry the sample and measure the weight of soil to determine the weight at the permanent wilting point (W_{PWP}).
 - c. Oven dry the same sample at 105°C for more than 30 h, which gives the dry weight of soil (W_d).
 - d. Determine the various moisture constants as per the procedure given in [Tables 17.7 and 17.8](#).

17.4 MEASUREMENT OF SOIL MOISTURE AT FIELD

The Field Scout TDR Series can also be used for soil moisture measurement at site. This instrument is portable and can accurately measure soil moisture across the full range of soil moisture conditions. The instrument can be employed for depth-integrated soil moisture measurement utilizing the three types (in length) of probe rods, viz., 7.5, 12, and 20 cm, to suit the desired soil depth. Other than these depths, it can be utilized to measure the moisture at deeper soil by installing the (polyvinyl chloride) PVC access tubes and take readings deeper in the soil profile.

The TDR 300 ([Fig. 17.6](#)) has two volumetric water content (VWC) modes: one for standard soils and one for higher clay soils. In VWC mode, the meter converts a measured electrical signal into percent soil moisture content using an equation valid over a wide range of mineral soils. In irrigation mode, the meter displays the relative water content on a scale of 0–100 corresponding to a user-defined upper and lower soil moisture reference level. The water deficit, the amount of water needed to bring the soil moisture content up to the upper reference level, is also calculated and displayed. The reference levels are easily programmed into the meter with accompanying software.

An internal data logger and RS-232 port allow for use with a (Differential Global Positioning System) DGPS for geo-referenced soil moisture measurements. The instrument also includes software and PC cable. The advantages of this instrument are the following:

1. Portability;
2. Measurement of soil moisture at field: no need for soil sampling and laboratory analysis;
3. Gives accurate readings;
4. Can be utilized for irrigation scheduling by continuous moisture readings;
5. Easy to calibrate;
6. Durable: more rugged probe block;
7. Easily change rod length selection with keypad on digital display

Table 17.7 Protocol for Determination of Soil Moisture Constants of the Field Plot

Field plot location:Lat,Long

Village:

Name or Outlet ID and RD:

Name of Minor/Distributary/Main canal:.....

Chak No.:.....

Khasra No.:.....

General Soil Characteristics: (i) Type:, (ii) Soil depth:.....

General Topography:.....

General Weather Condition during Sampling:

Date of Sampling:

Sample IDs:

Rooting Depth:cm

[illegible]

Soil sample depth: 20–50 cm															
1															
2															
3															
4															
Average															
Soil sample depth: 50–100 (=D) cm															
1															
2															
3															
4															
Average															
$V = \text{volume of sample (cm}^3\text{)} \left[V = \frac{\pi}{4}d^2h \right]; W_i = \text{initial weight of the sample taken at the time of sample (g).}$ BD, bulk density.															

Table 17.8 Determination of Average Value of Moisture Constants

[illegible]



FIGURE 17.6

Soil moisture measurement using the TDR 300.

17.5 CONCLUDING REMARKS

Soil type in the irrigation command area is the most important governing parameter for selecting the crop and water management practices as well as in deciding the irrigation application method. Water losses in the field are also different for different soil characteristics. Considering this, a detailed protocol for analyzing soil properties is discussed in detail in this chapter. The chapter also includes the various mechanisms of soil moisture assessment in the field itself using TDR 300 and laboratory, including fixing up of the soil-water constants, such as field capacity, permanent wilting point, and saturation moisture constant.

FURTHER READING

- IAEA, 2008. Field Estimation of Soil Water Content: A Practical Guide to Methods, Instrumentation, and Sensor Technology. International Atomic Energy Agency, Vienna.
- ISRIC, 2002. Procedures for Soil Analysis. International Soil Reference and Information System, Food and Agriculture Organization of United Nations, The Netherlands.
- MoAg, 2011. Methods Manual: Soil Testing in India. Department of Agriculture and Cooperation, Ministry of Agriculture, Government of India, New Delhi, India.
- Spectrum, 2012. Field Scout TDR 300 Soil Moisture Meter: Product Manual. Spectrum Technology Inc., Plainfield, IL, 60585.